

Question Paper

Exam Date & Time: 27-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Cloud Computing [IT_4441]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze and explain in detail about the economic benefits of cloud computing, specifically detailing the financial impact related to People, Hardware, Pay-as-you-go pricing, and Time to market. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) A large enterprise, Customer C1, contracts a mission-critical Cloud service from Provider P1. Mention the types of service level agreement available to choose. Also mention the objective parameters they can expect from the provider. (3)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) List down the uses of UDDI and WSDL foundational components in the Service-Oriented Architecture. (2)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) List out the merits of using MAX-MIN Fair allocation. Compute the max-min fair allocation for a set of four sources with demands 4, 2, 10, 4 and weights 2.5, 4, 0.5, 1 when the resource has a capacity of 16 (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) RIL Enterprises wants to use suitable compute virtualization for supporting their server operations. Suggest the types of hypervisor's available for compute virtualization with its advantages and explain. (3)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- C) Identify the scheduling objectives for batch systems in SOA? (2)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) A company is planning to migrate its critical applications to a virtualized data centre. What strategies should the company implement to avoid the single point of failure. Analyse how VM Fault Tolerance mechanisms are used. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) How VLAN trunking can be utilized for cloud environment and mention the benefits of VLAN (3)

trunking?

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Specify the compute service name in the Openstack and list the functions of it. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 4) Define Cloud Service Management and state its two primary objectives in relation to an organization's business goals and consumer expectations. Also, provide any three core management processes and briefly state the main goal of each chosen process. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) A financial services firm, 'GlobalTrust,' runs its high-frequency trading platform on a critical Virtual Machine (VM). Any downtime during trading hours is unacceptable and results in significant financial losses. Explain how the VM Fault Tolerance (VMFT) technique used for protecting compute resources and ensuring continuous business operations (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Why are microservices better than single monolithic applications for the cloud? How do they improve scalability and make applications more reliable? (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 5) Imagine a cybercriminal steals a copy of a critical Virtual Machine (VM) from a company's cloud. How do you define the security threat of VM theft, and what specific locks or restrictions must the cloud environment implement to ensure that the VM can only run on its original, trusted physical server? (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) In a cloud environment, how is a Demilitarized Zone (DMZ) used to create a secure buffer between the public internet and private systems? (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4441:Cloud Computing

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	6
3	3	CO 3	5
4	4	CO 4	19
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4441:Cloud Computing

No of

#	Level	Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	7	25(50%)
4	Analysis	7	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	14	50 (100%)